

# TARANG GUPTA

VIT University, Chennai | Roll No.: 23BEC1477

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## Education

VIT University, Chennai

Expected: May 2027

B.Tech, Electronics and Communication Engineering

CGPA: 8.97/10

St. Joseph's School, Shaktinagar — CBSE Class XII (9.0) | Class X (9.2), 2023 | 2021

## Technical Skills

**Languages & HDL:** C, Embedded C, Python, Verilog HDL, MATLAB, Assembly, Java

**Tools & EDA:** Synopsys Design Compiler, LTSpice, MATLAB & its Toolbox, OptiSystem, Git, Linux (Ubuntu), Cisco Packet Tracer, NumPy/SciPy, scikit-learn, TFLite

**Domains:** VLSI & SoC Design, RISC-V Architecture, Embedded Systems, Baseband DSP, MU-MIMO / IEEE 802.11be, RF Front-End Design, PolSAR Radar

## Projects

India RISC-V Chip Tapeout — Chip-to-Startup (C2S) Program

Apr 2025 – Oct 2025

Government of India

VIT Chennai / SCL 180 nm

- Selected for a nationally competitive government-backed tapeout program; completed RTL design and functional verification of core processor pipeline stages, synthesised in Synopsys Design Compiler on the SCL 180 nm node.

RIMUS — Autonomous Multi-Sensor Safety Rover

Jan 2026 – Apr 2026

Embedded C Programming Course Project

VIT Chennai

- Built an autonomous obstacle-avoiding rover on Arduino Mega: an IR bumper for instant emergency stops and a servo-scanned ultrasonic sensor for gradual slow-down, with active H-bridge braking to eliminate motor overshoot.
- Developed an ESP8266 sensor node hosting an embedded web server with a live JSON API, polling DHT11 and MQ-2 sensors with threshold-based hazard escalation.

EMG-Based Robotic Arm

Nov 2024

Embedded Systems Mini Project

VIT Chennai

- Designed a microcontroller-driven robotic arm mimicking hand motion from electromyographic signals: analog signal conditioning and amplification of raw EMG input, ADC-based digitisation, and real-time control logic driving servos for finger and wrist actuation.

PolarVerify — PolSAR-Based Ship Detection

Jan 2026 – Present

Radar Systems Course Project

VIT Chennai

- Designed a two-stage physics-guided detection pipeline combining Cloude–Pottier eigenvalue decomposition with CA-CFAR, using coherency-matrix features (entropy, anisotropy) as SVM inputs; strong clutter suppression on RADARSAT-2 quad-pol imagery, automated in Python.

## Internship

Industrial Intern

20 May – 20 June 2026

NTPC Limited — Singrauli Super Thermal Power Station (SSTPS)

Shaktinagar, Uttar Pradesh

- Worked in the Control & Instrumentation department of a 2,000 MW thermal station, studying the Honeywell DCS, marshalling cabinets routing field-instrument signals to the control room, and process interlock/alarm logic on plant control panels.
- Examined UPS, battery bank and static voltage & frequency converter systems providing uninterrupted power to critical C&I equipment, plus plant network architecture.

## Research & Publications

Wi-Fi 7 MU-MIMO Transceiver for Real-Time Aerial Swarm Coordination

Dec 2025 – Present

Co-author — IETE Journal of Research (Under Review)

VIT Chennai

- Simulated a 4×8 MU-MIMO downlink (IEEE 802.11be, 320 MHz, 4096-QAM) in MATLAB WLAN Toolbox; characterised BER, throughput and EVM across 10–50 dB SNR by Monte Carlo simulation, demonstrating 11.5 Gbps peak aggregate throughput at SNR > 42 dB.

Real-Time AI Audio Upscaling for Bandwidth-Constrained Wireless Links

June 2025 – Present

Co-author — IEEE ANTS 2026, IIT Roorkee (Submitted)

Samsung R&D, Bangalore

- Co-authored under Samsung PRISM: designed a deep learning pipeline (ATC-TFiLM, VCTK corpus) restoring speech quality from low-bitrate wireless inputs under real-time latency constraints, deployed on-device via TFLite.

## Certifications & Leadership

Embedded Systems Design — Maven Silicon (2026) | AI Engineering Professional Certificate — IBM (2026) | Logistics & Hospitality Lead, TEDxVITChennai (2024–25) | Knowledge Strategist, BBQC Quiz Club